Physics 5313: Electromagnetic Theory 2 (Fall 2026)

Lecture 14: Resonant Cavity

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Resonant cavity

- An enclosed volume used to trap and store standing waves.
- Electromagnetic standing waves in a volume bounded by perfectly conducting walls.
- Storage occurs only for waves that oscillate at one of a discrete set of frequencies because the source-free Maxwell equations and the field boundary conditions at the cavity walls define a normal-mode eigenvalue problem.

Example: Cylinder Resonant cavity

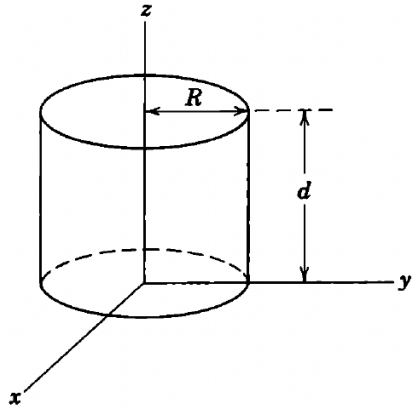
- Perfect conductor $e^{-i\omega t}$
- Governing Eq

$$\begin{cases} \nabla^2 \vec{E} + \mu\varepsilon\omega^2 \vec{E} = 0 \\ \nabla^2 \vec{H} + \mu\varepsilon\omega^2 \vec{H} = 0 \end{cases}$$

- Boundary condition

$$\vec{H}_\perp = 0, \quad \vec{E}_{//} = 0 \quad \text{on} \quad S$$

$$S: \begin{cases} r = R \\ z = 0, \ z = d \end{cases}$$



Example: Cylinder Resonant cavity

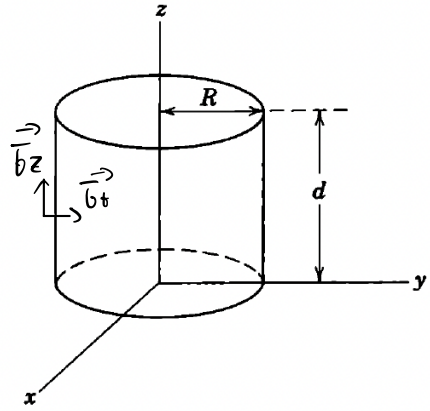
- Difference between cavity and waveguide

TM : $H_z = 0$

BC : $\vec{H}_t = 0$, $\vec{E}_{tan} = 0$ on S

$\Rightarrow$ $\begin{cases} r = R, & E_z = 0 \\ z = 0, d, & \vec{E}_t = 0 \end{cases}$

TE : $E_z = 0$ $\quad$ $\begin{cases} r = R, & \frac{\partial H_z}{\partial n} = 0 \\ z = 0, d, & H_z = 0, \ \vec{E}_t = 0 \end{cases}$



Example: Cylinder Resonant cavity

- Express transverse with z component

$$\textcircled{1}\quad \frac{\partial \mathbf{E}_t}{\partial z} + i\omega\hat{\mathbf{z}} \times \mathbf{B}_t = \nabla_t E_z, \qquad \textcircled{2}\ \hat{\mathbf{z}} \cdot (\nabla_t \times \mathbf{E}_t) = i\omega B_z$$

$$\textcircled{3}\ \frac{\partial \mathbf{B}_t}{\partial z} - i\mu\epsilon\omega\hat{\mathbf{z}} \times \mathbf{E}_t = \nabla_t B_z, \qquad \textcircled{4}\ \hat{\mathbf{z}} \cdot (\nabla_t \times \mathbf{B}_t) = -i\mu\epsilon\omega E_z$$

$$\textcircled{5}\quad \nabla_t \cdot \mathbf{E}_t = -\frac{\partial E_z}{\partial z}, \qquad \textcircled{6}\ \nabla_t \cdot \mathbf{B}_t = -\frac{\partial B_z}{\partial z}$$

Resonant cavity

- TE $E_z = 0$ We first find out H_z

$$\begin{cases} \nabla^2 H_z + \omega^2 \mu\varepsilon H_z = 0 \\ H_z = 0 \qquad z = 0, d \end{cases}$$

$$H_z = \psi(x,y) Z(z) e^{-i\omega t}$$

$$\nabla_t^2 \psi Z + \frac{\partial^2}{\partial z^2}(\psi Z) + \omega^2 \mu\varepsilon \psi Z = 0$$

Resonant cavity

$$\nabla_t^2(\psi Z) + \frac{\partial^2}{\partial z^2}(\psi Z) + \omega^2 \mu \varepsilon (\psi Z) = 0$$

- TE

$$\frac{1}{\psi}\nabla_t^2 \psi + \frac{1}{Z}\frac{\partial^2 Z}{\partial z} + \omega^2 \mu \varepsilon = 0$$

$$\| \\ -k^2$$

$$\begin{cases} \dfrac{\partial^2 Z}{\partial z^2} = -k^2 Z \\ Z = 0 \quad \text{on} \quad z = 0, d \end{cases} \Rightarrow Z = \sin\left(\frac{p\pi}{d} z\right) \qquad p = 1, 2, 3, \ldots$$

Resonant cavity

- TE $H_z = \psi(x,y)\sin\left(\frac{p\pi}{d}z\right)e^{-i\omega t}$

$$\begin{cases} \nabla_t^2 H_z + (\omega^2\mu\varepsilon - k^2)H_z = 0 \\ \frac{\partial H_z}{\partial n} = 0 \quad \text{at} \quad r = R \end{cases}$$

$$\psi = H_0 J_m(\gamma_{mn}\rho)e^{\pm im\rho}$$

Resonant cavity

$$\vec{B} = \mu \vec{H}$$

- TE

$$\text{①} \quad \frac{\partial \vec{E}_t}{\partial z} + i\omega \hat{z} \times \vec{B}_t = \nabla_t E_z = 0$$

$$\text{③} \quad \frac{\partial \vec{B}_t}{\partial z} - i\mu\varepsilon\omega \hat{z} \times \vec{E}_t = \nabla_t B_z$$

$$\text{①} \Rightarrow \quad \frac{\partial \vec{E}_t}{\partial z} = -i\omega\mu \hat{z} \times \vec{H}_t \qquad \text{④}$$

$$\frac{\partial}{\partial z}\,\text{③} \Rightarrow \quad \mu \frac{\partial^2 \vec{H}_t}{\partial z^2} - i\mu\varepsilon\omega \hat{z} \times \frac{\partial \vec{E}_t}{\partial z} = \mu \nabla_t \frac{\partial H_z}{\partial z}$$

Resonant cavity

- TE

$$k = \frac{p\pi}{d}$$

$$\mu \frac{\partial^2 \vec{H}_t}{\partial z^2} + i\omega\varepsilon \vec{z} \times (i\omega\mu \vec{z} \times \vec{H}_t) = \nabla_t \psi \cos(kz) = \mu k \cos(kz) \nabla_t \psi$$

$$\mu \frac{\partial^2 \vec{H}_t}{\partial z^2} + \omega^2 \mu \varepsilon \vec{H}_t = \nabla_t \psi \cos(kz) = \mu k \cos(kz) \nabla_t \psi$$

$$\textcircled{9} \quad \nabla_t \cdot \vec{H}_t = -\frac{\partial H_z}{\partial z} = -k \cos(kz) \nabla_t \psi$$

$$\vec{H}_t \propto \cos(kz)$$

Resonant cavity

- TE

$$(-\mu k^2 + \omega^2 \mu^2 \varepsilon)\, H_f^- = \mu k \cos(kz)\, J_f \psi$$

$$H_f^- = \frac{\mu k}{\omega^2 \mu^2 \varepsilon - \mu k^2} \cos(kz)\, J_f \psi$$

Introduce $\gamma^2 = \omega^2 \mu \varepsilon - \left(\frac{p\pi}{d}\right)^2$

$$H_f^- = \frac{pz/d}{\gamma^2} \cos\left(\frac{p\pi}{d} z\right) J_f \psi$$

Resonant cavity

- TE

$$\frac{\partial \vec{b}_t}{\partial z} = -i\omega\mu \hat{z} \times \vec{H}_t \qquad (a)$$

$$= -i\omega\mu \frac{P\pi/d}{\gamma^2} \cos\left(\frac{P\pi}{d} z\right) \left(\hat{z} \times \nabla_t \psi\right)$$

$$\Rightarrow \quad \vec{b}_t = -\frac{i\mu\omega}{\gamma^2} \sin\left(\frac{P\pi}{d} z\right) \left(\hat{z} \times \nabla_t \psi\right)$$

Resonant cavity

- TE

$$H_z = \psi(x,y) \sin\left(\frac{p\pi}{d} z\right) e^{-i\omega t}$$

$$\vec{E}_t = -\frac{i\omega\mu}{\gamma^2} \sin\left(\frac{p\pi}{d} z\right) (\hat{z} \times \nabla_t \psi)\, e^{-i\omega t}$$

$$\vec{H}_t = \frac{p\pi/d}{\gamma^2} \cos\left(\frac{p\pi}{d} z\right) \nabla_t \psi\, e^{-i\omega t}$$

- TM

$$E_z = \psi(x,y) \cos\left(\frac{p\pi}{d} z\right) e^{-i\omega t}$$

$$\vec{E}_t = -\frac{p\pi}{d\gamma^2} \sin\left(\frac{p\pi z}{d}\right) \nabla_t \psi\, e^{-i\omega t}$$

$$\vec{H}_t = \frac{i\varepsilon\omega}{\gamma^2} \cos\left(\frac{p\pi}{d} z\right) (\hat{z} \times \nabla_t \psi)\, e^{-i\omega t}$$

Resonant frequency

$$\gamma^2 = \mu\varepsilon\omega^2 - k^2$$

$$\gamma_\lambda^2 = \mu\varepsilon\omega^2 - \left(\frac{p\pi}{d}\right)^2$$

$$\omega_\lambda^2 = \frac{1}{\mu\varepsilon}\left[\gamma_\lambda^2 + \left(\frac{p\pi}{d}\right)^2\right]$$

TM mode frequency

$\bar{E}_z = 0$ at $\rho = R$

$$\psi(\rho, \phi) = \bar{E}_0 \bar{J}_m(\gamma_{mn}\rho)\, e^{\pm im\varphi}$$

$$\gamma_{mn} = \frac{X_{mn}}{R}$$

X_{mn} is the nth root of $\bar{J}_m(x) = 0$

$m = 0, 1, 2, \cdots$ $n = 1, 2, 3, \cdots$

$$W_{mnp} = \frac{1}{\sqrt{\mu\epsilon}} \left(\frac{X_{mn}^2}{R^2} + \frac{p^2\pi^2}{d^2} \right)^{1/2}$$

Lowest: $m = 0$, $n = 1$, $p = 0$ $\quad W_{TM010} = \dfrac{2.405}{\sqrt{\mu\epsilon}\, R}$

TE mode frequency

$$\frac{\partial H_z}{\partial n} = 0 \quad \text{at} \quad \rho = R$$

$$\psi(\rho, \varphi) = H_0 J_m(\gamma_{mn}\rho) e^{\pm im\varphi}$$

$$\gamma_{mn} = \frac{x'_{mn}}{R}$$

x'_{mn} is the nth root of $J'_m(x) = 0$

$m = 0, 1, 2 \cdots \quad n = 1, 2, 3 \cdots \quad p = 1, 2, 3 \cdots$

TE mode frequency

Smallest root for $J_m'(x)=0$ is $m=1, n=1$

instead of $m=0, n=1$

$m=1,\ n=1,\ p=1$

$$\omega_{TE111} = \frac{1.841}{\sqrt{\mu\varepsilon}\,R}\left(1+2.912\frac{R^2}{d^2}\right)^{1/2}$$

When $d > 2.03R$, $\omega_{TE111} < \omega_{TM010}$